

Accelerating MATSim-Based Vertiport Siting via Facility Location Surrogate Space Mapping

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1 Introduction

Urban Air Mobility (UAM) using electric vertical take-off and landing (eVTOL) aircraft is emerging as a candidate passenger mode, with market projections reaching USD 210 billion by 2045 [1]. Its viability, however, hinges on infrastructure: the location of vertiports largely determines whether a UAM network is operationally and economically viable. Because siting interacts with travellers' mode and route choices, access and egress, and fleet operations (all emergent from individual behaviour), any candidate network must be assessed with an agent-based simulation (ABS) such as MATSim-UAM [2]. To optimise the network, current studies either embed MATSim-UAM directly inside a bi-level metaheuristic search [3], which requires thousands of costly runs, or replace it with a data-fitting surrogate (e.g., neural network, Gaussian process), whose training itself consumes a large corpus of simulation runs while offering little physical insight [4, 5]. We propose a Facility Location Model-based Space Mapping (FLM-SM) framework that couples MATSim-UAM with a mechanistic, reduced-order surrogate: a facility location model (FLM) solved to optimality and governed by a small set of interpretable parameters. These parameters are iteratively calibrated so that the FLM's siting response aligns with MATSim-UAM, letting the network search run cheaply on the surrogate while the full simulation is invoked only to correct the mapping and evaluate the siting result. The framework is demonstrated on the Corsica and Munich scenarios and benchmarked for efficiency against a bi-level genetic algorithm (GA) optimisation. Experiments show that, on Corsica, the framework attains 96% of the GA baseline's objective value while using only 3.3% of its computational budget, with the remaining 4% gap falling within the run-to-run stochastic variability of the agent-based simulation.

2 Methodology

Like most previous studies, this study focuses on the classic vertiport location problem of minimising the number of vertiports while maximising UAM travel demand. The set of candidate vertiport sites is obtained by applying k-means clustering to the population distribution, yielding 50 candidate locations from which the model selects the active vertiports. To describe the siting problem, an FLM that maximises UAM demand is first developed. The total UAM demand is given by

$$\max_{\mathbf{y}} D^{UAM} = \sum_{i,i'} \sum_{k \in K_{ii'}^{UAM}} d_{ii'k} \quad (1)$$

which sums the route-level demand $d_{ii'k}$ over all origin-destination (OD) pairs (i, i') and candidate UAM routes $k \in K_{ii'}^{UAM}$, where $K_{ii'}^{UAM}$ is the set of candidate UAM routes for the pair and the binary vector $\mathbf{y}$ marks the open vertiports. Each $d_{ii'k}$ follows a multinomial logit:

$$d_{ii'k} = d_{ii'}^{\max} \cdot \frac{a_{ii'k}(\mathbf{y}) \exp(V_{ii'k})}{\sum_{k' \in K_{ii'}^{UAM}} a_{ii'k'}(\mathbf{y}) \exp(V_{ii'k'}) + \sum_{g \in K_{ii'}^{ground}} \exp(V_{ii'g})} \quad (2)$$

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where $d_{ii'}^{\max}$ denotes the potential demand of OD pair (i, i') , and $K_{ii'}^{\text{ground}}$ collects the competing ground alternatives g . The availability term $a_{ii'k}(\mathbf{y}) = y_j y_{j'}$ equals one only when both endpoint vertiports of route $k = (j, j')$ are open, which is how the siting decision $\mathbf{y}$ enters the modal split. The utility adopts the same linear specification as the DiscreteModeChoice (DMC) module configured in MATSim [6], ensuring the coarse model shares an identical choice kernel with the fine model:

$$V_{ii'k} = \alpha_k + \sum_s \beta_{\text{time},k,s} (t_{k,s} + \theta_{k,s}) + \beta_{\text{cost}} \left(\frac{\text{dist}_{ii'}}{\text{dist}_{\text{ref}}} \right)^\lambda c_k \quad (3)$$

where α_k is the alternative-specific constant, s indexes the legs of route k , $t_{k,s}$ and c_k are the segment time and route cost, $\beta_{\text{time},k,s}$ and β_{cost} are the time and cost coefficients, $\text{dist}_{ii'}$ is the OD distance normalised by the reference dist_{ref} with exponent λ , and $\theta_{k,s}$ is the mapping correction applied to the ground segments to inject the MATSim-observed access, egress, and ground travel times into the coarse model. The correction θ is stored as a matrix over space and time, indexed by network location and by hour of day, so that it captures the hourly road-network congestion delay applied to each ground segment. At this point, the coarse model is fully and explicitly specified, embedding the road network information. The route-choice probabilities can be evaluated as constants once the correction term θ has been mapped from the actual MATSim evaluation results, and the bilinear availability terms are linearised, so that the model reduces to a mixed-integer linear programming (MILP) solvable by Gurobi.

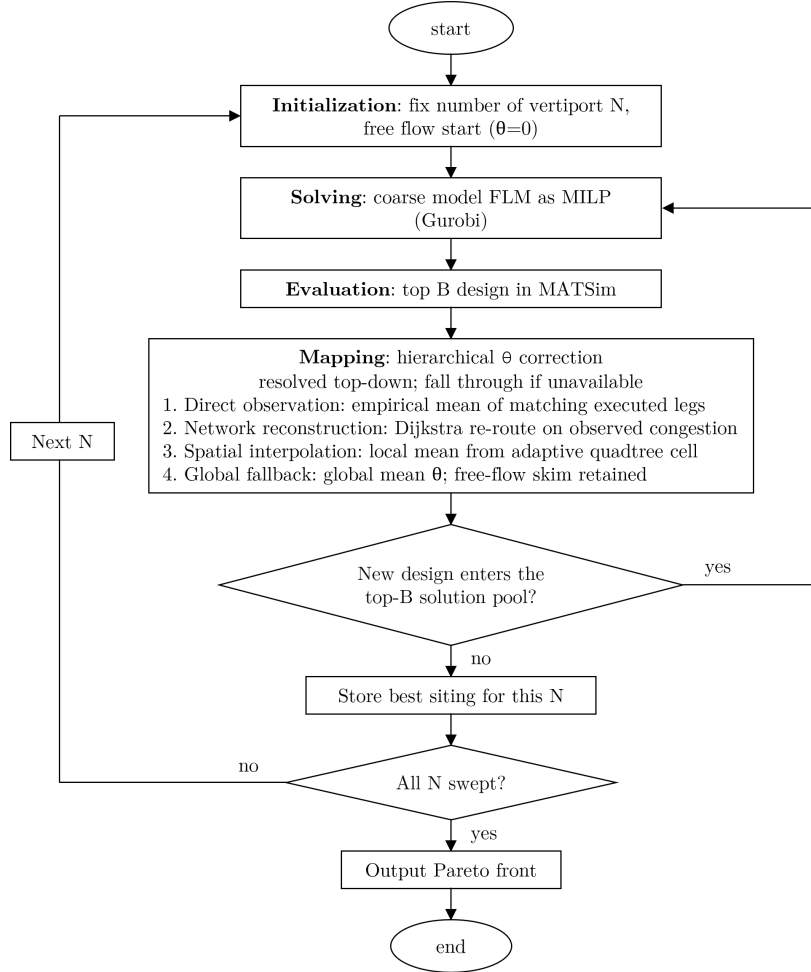


Figure 1: Facility-Location Surrogate Space Mapping Algorithm

The FLM is constructed as a coarse surrogate for the computationally expensive MATSim simulation, following a space-mapping scheme as shown in Fig. 1. Starting from a free-flow assumption ($\theta = 0$), the surrogate MILP is solved by Gurobi, and the top- B designs retained in its solution pool (here $B = 6$: the proven optimum together with the next-best feasible sitings) are evaluated in a single batch by the high-fidelity MATSim fine model. From the executed legs, the discrepancies in congestion-related car travel time and access/egress time are extracted and injected back into the surrogate as a correction term θ . This correction is resolved through a four-tier priority cascade: for each origin–destination or leg key, a directly

observed empirical mean is used when available; otherwise the value is reconstructed by Dijkstra re-routing over the observed link congestion, then interpolated from the enclosing cell of an adaptive quadtree built on accumulated observations, and finally assigned a global fallback. The corrected surrogate is re-solved, and the loop repeats while new designs continue to enter the top- B pool, with a monotone best-so-far retention guaranteeing no-regret convergence to the returned siting. Through this iterative refinement, the surrogate progressively aligns with the real system dynamics, substantially reducing the number of costly simulations required.

3 Results

The efficiency and effectiveness of FLM-SM are evaluated against NSGA-II (Non-dominated Sorting Genetic Algorithm II) and GA baseline methods on the Corsica and Munich scenarios, with the mode-choice utility specification and parameters following Fu et al. and Hörl [7, 8]. Since DMC does not rely on “innovation-memory” convergence, 10 MATSim iterations suffice for the evaluation to stabilise, and the average of the last 6 is taken as the evaluation; the same 10-iteration budget is also adopted by [3]. In terms of search budget, NSGA-II is run for 300 true evaluations in total (population 10, 30 generations), whereas the GA baseline is run per vertiport-count target at two budgets, 200 evaluations (population 10, 20 generations) and 48 evaluations (population 8, 6 generations). Both the GA baseline and FLM-SM are single-objective: they maximise UAM demand under a fixed vertiport-count target, which is swept across targets to trace the front (an ϵ -constraint scheme), whereas NSGA-II is genuinely multi-objective. Hypervolume (HV) is normalised to the combined ideal-to-nadir box, evaluated on the common set of vertiport-count targets produced by the GA baseline; results are summarised in Table 1.

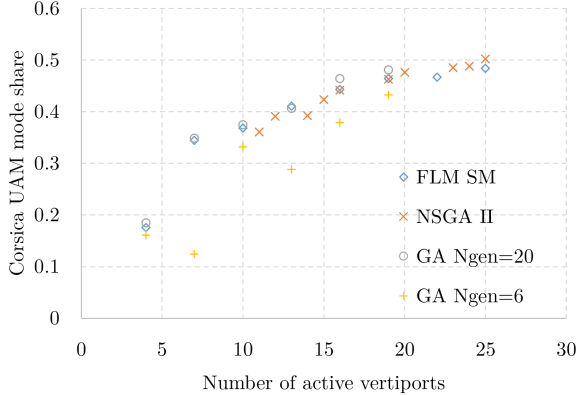


Figure 2: Pareto of Corsica

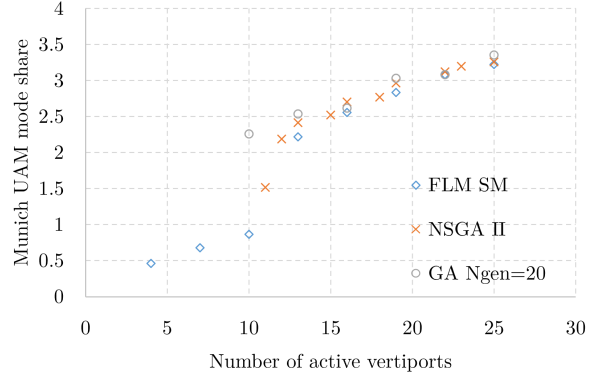


Figure 3: Pareto of Munich

Table 1: Comparison of optimisation methods across scenarios. Metrics are evaluated on the common set of vertiport-count targets produced by the GA baseline in each scenario. HV is computed on objectives normalised to the combined ideal-to-nadir box with reference point (1.1, 1.1).

Method	Corsica			Munich		
	HV	Evals	Evals/pt.	HV	Evals	Evals/pt.
FLM-SM	0.771	39	6.5	0.785	45	7.5
GA-Ngen20	0.800	1200	200	0.949	1200	200

On Corsica (Fig. 2), the four fronts overlap closely. The 20-generation GA attains the highest normalised hypervolume (0.800), marginally ahead of FLM-SM (0.771), with both lying above NSGA-II and the coarse-budget GA. FLM-SM retains the non-dominated solution at 13 vertiports. FLM-SM reaches this near-best front at roughly 3.3% of the true-evaluation cost of the 20-generation GA (6.5 versus 200 evaluations per Pareto point), while certifying each optimisation as a global MILP optimum rather than a heuristic non-dominated solution.

On Munich (Fig. 3), the 20-generation GA has a predominant result; FLM-SM and GA did not provide many dominating solutions. Fronts of FLM-SM and NSGA-II are effectively indistinguishable (the residual gap at 22 and 25 vertiports, 1.3% and 1.1%, is within the fluctuation expected from re-seeding), and NSGA-II attains a marginally higher normalised hypervolume (0.707 versus 0.665). At low vertiport counts, however, FLM-SM is clearly inferior: around ten active vertiports it substantially underestimates attainable demand,

reaching only 0.86% mode share against 1.52% for the nearest NSGA-II solution, and the FLM-SM front rises steeply thereafter (a $2.6\times$ jump between 10 and 13 vertiports). We attribute this shortfall to the coarse model: in a sparse-network, low-demand regime the surrogate cannot fully reproduce the congestion dynamics and mode-choice adjustments that emerge from the agent-based MATSim simulation, so the correction terms alone do not recover the demand that the fine model realises. This limitation is scenario-dependent and most pronounced when the UAM network is sparse; as coverage increases and the two methods’ budgets overlap, the discrepancy vanishes into seed-level noise.

4 Discussion

Figures 4 and 5 overlay two spatial diagnostics on the B16 and B25 Corsica configurations: the log-scaled heatmap of realised UAM access/egress road usage, and, per quadtree cell, the share of origin trips for which UAM is genuinely time-competitive with the car. The travel-time-savings ratio is defined as $r_{tts} = 1 - t_{uam}/t_{car}$, so a positive value means UAM is faster; the greyscale reports, among all car-and-UAM trip pairs originating in a cell, the fraction with $r_{tts} > 0$, while cells with fewer than ten eligible pairs are left unshaded so that the annotated per-cell counts separate genuinely low-competitiveness cells from under-sampled ones. Read together, the two layers show where access and egress loading concentrates onto a limited set of road segments, precisely the segments on which the ground-segment corrections θ carry the most weight, and thus an empirical motivation for confining the space-mapping correction to the access, egress, and ground-travel legs while leaving the flight segment uncorrected.

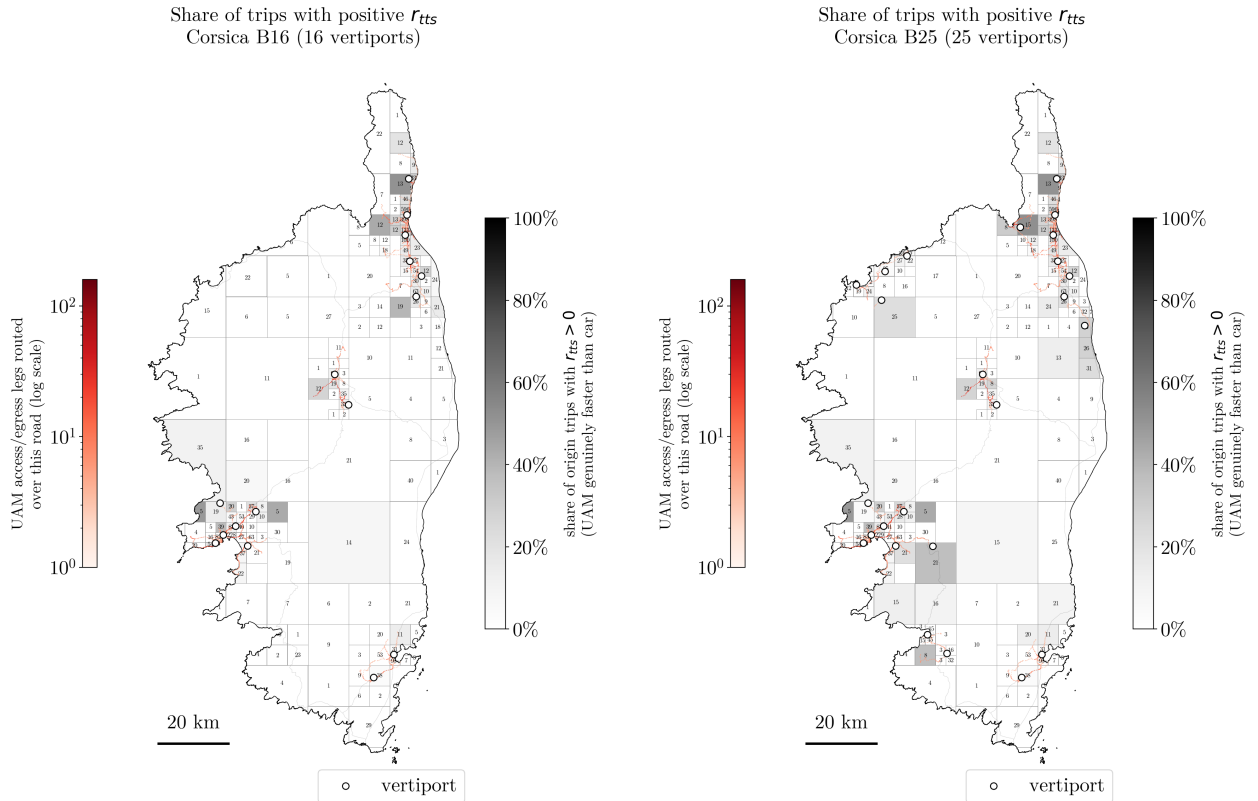


Figure 4: RTTS map of Corsica (16 vertiports activated)

Figure 5: RTTS map of Corsica (25 vertiports activated)

Both Corsica configurations exhibit a strongly corridor-bound competitiveness. UAM time advantages, and the access/egress traffic that serves them, concentrate in the north-eastern Bastia corridor, with a secondary cluster around Ajaccio, while the mountainous interior and the south remain largely unshaded. Expanding the network from 16 to 25 vertiports extends coverage into the north-western Balagne coast and the southern tip, but the added cells are predominantly light and thinly sampled: the additional vertiports broaden geographic reach without creating new high-competitiveness regions. This spatial saturation mirrors the diminishing marginal demand along the Pareto front and locates its cause in the demand geography: beyond the few corridors where terrain forces long road detours, UAM’s straight-line advantage is not large enough to overturn the car.

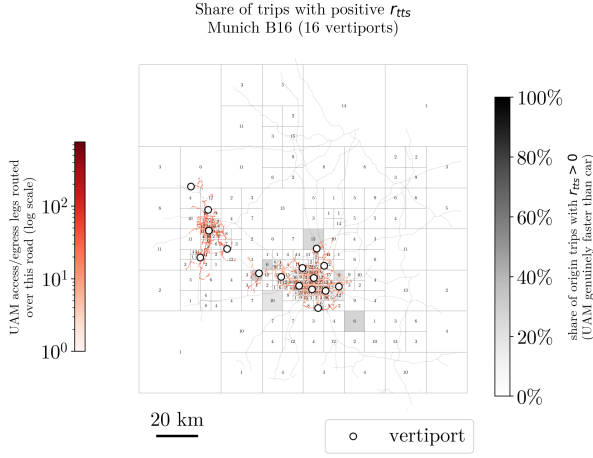


Figure 6: RTTS map of Munich (16 vertiports activated)

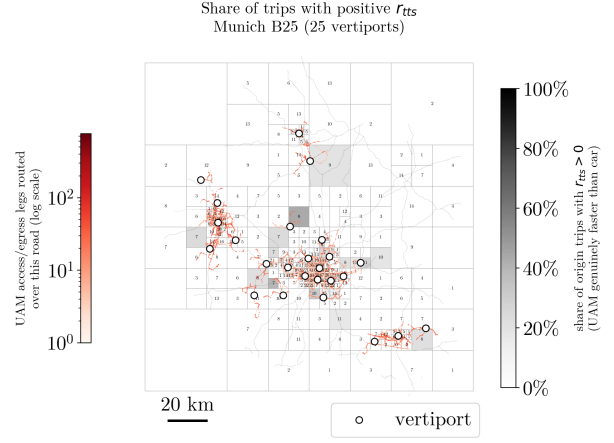


Figure 7: RTTS map of Munich (25 vertiports activated)

The Munich configurations (Figs. 6 and 7) present a markedly different spatial signature. Even in the dense central cluster the share of origin trips for which UAM is time-competitive rarely exceeds forty per cent, and none of the near-saturated shading seen along the Corsican Bastia corridor appears here. Access and egress loading concentrates tightly on the central Munich area, with one detached western cluster, while the vertiports added between B16 and B25 attract only thin peripheral traffic. Consistent with this, the quadtree resolves the centre into many small, densely sampled cells against a sparsely populated periphery: a high but strongly centralised demand base, rather than the corridor-bound pattern of Corsica, which is consistent with Rothfeld’s previous studies on airport distribution [2].

This contrast clarifies the aggregate Munich results. The high absolute UAM mode share observed there reflects the large underlying trip volume of a major metropolitan region, not superior time competitiveness: on level terrain served by a fast, well-connected road network, UAM’s straight-line advantage is largely offset, and the fixed access, egress, and terminal-handling times weigh heavily on the short intra-urban trips that dominate the centre. Time-competitive origins are therefore confined to a compact core, which explains why the low-budget configurations, unable to cover that core, sharply underestimate attainable demand, and why at higher vertiport counts the competing methods converge to within seed-level noise, as additional vertiports spread the same pale, low-competitiveness coverage more widely without unlocking new advantaged regions.

5 Conclusion

This study presents an FLM-based Space Mapping framework that accelerates MATSim-UAM vertiport siting by replacing repeated high-fidelity optimisation with a mechanistic MILP surrogate. Results show that FLM-SM can achieve performance close to GA and NSGA-II while substantially reducing the number of costly MATSim evaluations. The results also indicate that its accuracy is scenario-dependent, with larger discrepancies occurring in sparse UAM networks. Overall, FLM-SM provides an efficient and interpretable alternative for simulation-based vertiport network optimisation. Nevertheless, FLM-SM still holds potential value as an efficient and interpretable framework for simulation-based optimisation, particularly through its globally optimal MILP solutions and physically meaningful intermediate variables. Future work will focus on further improving the FLM-SM model through a deeper investigation of MATSim-DMC behaviour and congestion mechanisms, and extending the current FLM to a capacitated facility location model (CFLM) to account for vertiport capacity constraints.

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